PRAGYA AI

PRODUCT REQUIREMENTS DOCUMENT

Bridging Indian Knowledge Systems with Modern Science

Version 1.0 | November 2026

A multilingual, multimodal AI knowledge and learning platform with specialized personas and a verified Sanskrit evidence system.

1. Product Overview

Product Name: Pragya AI

Product Type: Multilingual, multimodal, AI-powered knowledge and learning platform.

Project Theme: Indian Knowledge System (IKS) & Modern Science.

Platform: Responsive web application for desktop, tablet, and mobile.

Pragya AI connects knowledge found in ancient Indian texts and traditions with modern scientific concepts, mathematics, business strategy, and psychology. It offers four specialized AI personas, supports text and image-based conversations in English, Hindi, and Hinglish, and provides a verified Sanskrit evidence system wherever reliable references are available.

2. Problem Statement

Indian literature contains extensive ideas relating to philosophy, mathematics, astronomy, governance, psychology, and other disciplines. However, accessing, understanding, and relating this knowledge to modern fields can be challenging.

- General-purpose AI assistants are not specifically designed to connect Indian Knowledge Systems with modern disciplines.
- AI systems may generate incorrect or unverifiable Sanskrit verses and citations.
- Historical knowledge, philosophical interpretation, and established scientific evidence are not always clearly distinguished.
- Users may need separate tools for science, mathematics, business strategy, and psychological guidance.
- Language barriers can make traditional knowledge less accessible.

Product response: Pragya AI addresses these challenges through specialized AI personas, multilingual interaction, and a verified knowledge-reference system.

3. Product Objectives

- Make Indian Knowledge Systems accessible through modern AI technology.
- Connect ancient Indian texts and concepts with relevant modern disciplines.
- Provide specialized assistance through four dedicated AI personas.
- Support English, Hindi, and Hinglish communication.
- Provide reliable Sanskrit references with source and verification information.
- Allow users to maintain personal accounts and access previous conversations.
- Support image uploads and AI-powered image understanding.
- Deliver a secure, responsive, and user-friendly experience.

4. Target Audience

User group	Primary use case
Students	Learn and explore connections between ancient and modern knowledge.
Researchers	Explore Indian texts and their interpretations.
Educators	Explain traditional concepts through modern examples.
Entrepreneurs	Apply traditional management and governance principles.
Mathematics and CS learners	Explore mathematical concepts and algorithms.
General users	Understand philosophy, psychology, and Indian knowledge.

5. Core Product Features

5.1 User Authentication and Account Management

- Google sign-in and account registration.
- Secure session management and logout.
- Automatic profile creation after registration.
- Display user name, email, and profile picture.
- Account deletion and user-specific data isolation.

5.2 Personal User Profile

- View basic account information.
- Store individual user preferences.
- Maintain user-specific chat history and settings.
- Prevent access to another user's private information.

5.3 AI Persona System

Persona	Domain and capabilities	AI provider
Pragya-Veda	Science and cosmology: connects relevant Vedic and Upanishadic concepts with physics, astronomy, and anatomy; distinguishes traditional interpretations from established scientific evidence.	Google Gemini
Pragya-Neeti	Corporate strategy: business and startup guidance, leadership, governance, decision-making, and management informed by Arthashastra and Vidur Niti.	Google Gemini
Pragya-Ganita	Mathematics and computer science: mathematical problem-solving, Vedic Mathematics, modern algorithms, and computational complexity such as Big-O.	Groq
Pragya-Sarathi	Psychology and wellness: general emotional support, CBT-oriented concepts, reflection, mindfulness, and relevant teachings from the Bhagavad Gita and Patanjali Yoga Sutras.	Google Gemini

- Each persona has a dedicated system prompt and domain boundaries.
- The selected persona remains associated with its conversation.
- Backend configuration controls provider selection.
- Users can start new conversations with different personas.

- The AI communicates when a question is outside its expertise.

5.4 New Chat and Conversation Management

- Create a new conversation and assign a persona.
- Generate a title from the initial user message.
- Save messages in chronological order and update conversation timestamps.
- Rename, delete, reopen, and continue previous conversations.
- Preserve relevant context when continuing a conversation.

5.5 Recent Chat History

- Display recent conversations, sorted by last update.
- Show conversation titles and associated personas.
- Open, rename, or delete conversations.
- Show an appropriate empty state when no history exists.

5.6 Search Previous Conversations

- Search conversation titles and message content.
- Display matching conversations and allow users to open them.
- Restrict results to the authenticated user.
- Support English initially, with Hindi and Hinglish search improvements through text normalization and indexing.

5.7 Multimodal Chat Interface

- Text-based conversations and image attachments.
- Message sending, receiving, loading indicators, and error/retry states.
- Distinct user and AI messages.
- Structured AI responses and Sanskrit Evidence Boxes where relevant.
- Support long conversations and message scrolling.

5.8 Image Upload and AI Image Understanding

- Upload JPEG, JPG, PNG, and WEBP images; initial limit: 5 MB per image.
- Preview and remove an image before submission.
- Store uploads securely and associate them with messages.
- Send actual image content to a compatible vision-capable model.
- Support questions about mathematical problems, Sanskrit manuscripts, scientific diagrams, and business charts.

Important: Storing an image URL alone does not enable image understanding; the image content must be processed by a vision-capable model.

5.9 Multilingual Communication

Language	Requirement
English	Understand and respond in English.
Hindi	Understand and respond in Hindi.
Hinglish	Understand mixed Hindi-English and respond naturally.

- Detect input language and respond in the same language by default.
- Allow language preferences in settings and language switching during conversations.
- Preserve technical terminology where appropriate.
- Display original Sanskrit verses in Devanagari script regardless of conversation language.

5.10 Sanskrit Evidence System

Core differentiator: Relevant answers should include an Evidence Box containing the granth/source name, chapter and verse number where available, original Sanskrit in Devanagari, Hindi and English translations, source or edition, and verification status.

- AI-generated citations must never be treated as proof.
- References must be retrieved from the verified knowledge database.
- Only records marked verified may be presented as verified.
- Partially verified references must be labelled clearly.
- If no reliable reference exists, display: “No verified reference found.”
- Never invent verses, chapter numbers, or source details.
- Distinguish interpretations from literal translations.
- Do not force a Sanskrit reference when no relevant source exists.

5.11 User Settings and Language Preferences

- Preferred language: English, Hindi, or Hinglish.
- Theme: Light, Dark, or System.
- Default AI persona.
- Chat-history saving preference.
- Persist preferences between sessions.

If chat-history saving is disabled, the application must clearly explain whether new conversations are ephemeral; existing history should not be deleted unless the user explicitly deletes it.

5.12 Responsive Dashboard

- Responsive desktop, tablet, and mobile layouts.
- Responsive sidebar and mobile navigation.
- Properly scaled text, images, and chat workspace.
- Readable typography and accessible controls.

6. User Journey

1. Open Pragya AI.

2. Sign in using Google.
3. Enter the personalized dashboard.
4. Select Pragya-Veda, Pragya-Neeti, Pragya-Ganita, or Pragya-Sarathi.
5. Submit a text prompt or upload an image.
6. Backend authenticates the user, loads context, retrieves relevant evidence, and routes the request to the configured AI provider.
7. The selected persona generates an answer in the appropriate language.
8. Relevant verified Sanskrit references are displayed.
9. The conversation is saved and can be reopened, subject to the user's history-saving preference.

7. Technical Architecture

7.1 Proposed Technology Stack

Layer	Technology	Purpose
Frontend	Next.js	Web application
Styling	Tailwind CSS	Responsive UI
UI components	Shadcn UI	Reusable components
Icons	Lucide React	Interface icons
Backend	Supabase Edge Functions	Server-side application logic
Database	Supabase PostgreSQL	User data, conversations, and evidence
Authentication	Supabase Auth	Google authentication
File storage	Supabase Storage	Secure image storage
AI – Veda	Google Gemini	Science and cosmology
AI – Neeti	Google Gemini	Business and strategy
AI – Ganita	Groq	Mathematics and computer science
AI – Sarathi	Google Gemini	Psychology and wellness
Version control	GitHub	Source code management
Deployment	Vercel	Frontend hosting

7.2 High-Level Request Workflow

1. User interacts with the Next.js frontend.
2. Supabase Auth verifies the user's session.
3. Frontend sends the request to a secure backend Edge Function.
4. Backend validates user, conversation ownership, and selected persona.
5. Relevant conversation history is retrieved.
6. Evidence database is queried for relevant verified references.
7. Backend routes the request to Gemini or Groq.
8. AI generates a response.

9. Backend validates evidence identifiers against database records.

10. Assistant response and evidence associations are saved.

11. Frontend displays the response and Evidence Box.

Security: AI API keys must remain in server-side environment secrets and must never be exposed in frontend code.

8. Database Architecture

Supabase PostgreSQL will be the primary application database. Supabase Auth manages authentication independently. The application will maintain eight primary application tables.

No.	Table	Purpose
1	profiles	User profile information
2	user_settings	User preferences and settings
3	conversations	Chat sessions and persona association
4	messages	User and AI messages
5	attachments	Uploaded image metadata
6	evidence_sources	Source books, editions, and publications
7	evidence_verses	Sanskrit verses and translations
8	message_evidence	Links AI messages to evidence records

8.1 Key Relationships

- Auth users → profiles (one-to-one).
- Auth users → user settings (one-to-one).
- Auth users → conversations (one-to-many).
- Conversations → messages (one-to-many).
- Messages → attachments (one-to-many).
- Evidence sources → evidence verses (one-to-many).
- Messages ↔ evidence verses through message_evidence (many-to-many).

8.2 Storage Architecture

- user-uploads: private user-uploaded images.
- user-avatars: profile pictures; private by default.
- public-assets: public application assets.
- Use a user-scoped storage path such as {user_id}/{conversation_id}/{filename}, and enforce ownership through Storage policies.

8.3 Database Security

- Enable Row Level Security (RLS) on user-specific tables.
- Restrict private data access to the authenticated owner.

- Prevent ordinary users from modifying verified evidence records.
- Apply appropriate Storage access policies.
- Keep AI credentials in backend secrets.
- Validate backend requests and clean up personal records and stored files during account deletion.

9. Backend Functional Requirements

Operation	Responsibility
chat	Process user messages and route AI requests.
generate-title	Generate conversation titles.
search-conversations	Search personal chat history.
delete-conversation	Delete a conversation and related data.
delete-account	Coordinate account and user-data deletion.

Simple operations such as retrieving settings or listing conversations can use the Supabase client with appropriate RLS policies instead of requiring a separate Edge Function.

9.1 AI Request Processing

1. Authenticate the user.
2. Validate the request.
3. Verify conversation ownership.
4. Identify the selected persona.
5. Load relevant conversation history.
6. Retrieve relevant evidence records.
7. Apply the appropriate persona system prompt.
8. Route the request to the correct AI provider.
9. Handle text and image input.
10. Validate returned evidence identifiers.
11. Save the AI response.
12. Return the structured response to the frontend.

The frontend must not be trusted to determine the provider or supply authoritative system prompts.

10. Non-Functional Requirements

Category	Requirement
Performance	Fast dashboard loading and responsive interactions.
Security	Authentication, RLS, and secure API key management.
Privacy	Private user conversations and uploads.
Reliability	Graceful handling of AI provider errors.
Scalability	Architecture that can support more users and personas.
Availability	Clear error states and retry mechanisms.
Usability	Simple, intuitive interface.
Accessibility	Readable typography and accessible controls.
Compatibility	Modern desktop and mobile browsers.
Maintainability	Modular components and organized backend logic.
Accuracy	Verified references and transparent uncertainty.

11. AI Safety and Content Integrity

- Never fabricate Sanskrit verses or source references.
- Clearly distinguish traditional knowledge from modern scientific evidence.
- Do not claim that an ancient text scientifically proves a modern theory without reliable evidence.
- Communicate uncertainty where information is disputed or incomplete.
- Do not present interpretations as literal translations.
- Avoid misleading medical, financial, legal, or psychological claims.
- Pragya-Sarathi must not claim to replace professional mental-health care.
- Handle sensitive user information responsibly.
- Refuse or safely redirect harmful requests.

12. Scope and Feature Prioritization

No.	Feature	Priority
1	Google Sign In / Sign Up	Must Have
2	Personal User Profile	Must Have
3	New Chat	Must Have
4	Recent Chat History	Must Have
5	Search Previous Conversations	Should Have
6	Rename and Delete Chats	Must Have
7	Four AI Personas	Must Have
8	Image Upload in Chat	Must Have
9	AI Image Understanding	Must Have
10	Hindi, English, and Hinglish Support	Must Have
11	Sanskrit Evidence Box	Must Have
12	User Settings and Language Preference	Should Have
13	Responsive Mobile and Desktop Dashboard	Must Have

13. Development Roadmap

1. Phase 1 – Foundation and Database: Create Supabase project; configure tables, relationships, indexes, RLS, Storage buckets, and policies.
2. Phase 2 – Authentication and Profiles: Configure Google OAuth; implement login/logout, profiles, and settings.
3. Phase 3 – Frontend Dashboard: Build responsive dashboard, sidebar, persona selector, chat workspace, and mobile navigation.
4. Phase 4 – Chat Management: Implement conversations, message persistence, history, search, rename, and delete.
5. Phase 5 – AI Integration: Configure Gemini and Groq; develop persona prompts, backend routing, and context handling.
6. Phase 6 – Sanskrit Evidence System: Build verified dataset, source and verse records, retrieval, validation, and Evidence Box UI.
7. Phase 7 – Multimodal Capabilities: Implement image upload, private storage, vision-model processing, and image-question testing.
8. Phase 8 – Testing and Deployment: Test personas, languages, evidence accuracy, RLS, responsiveness, and deployment.

14. Acceptance Criteria

- Users can register and log in using Google.
- Every user has an independent profile.
- Users can create new conversations.
- All four personas work within their assigned domains.
- Each persona routes requests to its configured AI provider.

- Conversations are saved and can be reopened.
- Users can search, rename, and delete conversations.
- Users can upload supported images and receive image-aware answers.
- English, Hindi, and Hinglish conversations work.
- Relevant verified Sanskrit references appear in Evidence Boxes.
- Unverified references are never presented as verified.
- User preferences persist between sessions.
- The dashboard works on desktop, tablet, and mobile.
- RLS and Storage policies prevent unauthorized access.
- API keys are not exposed in frontend code.

15. Future Enhancements

- Voice input and AI voice responses.
- Real-time multilingual voice conversations.
- Sanskrit verse pronunciation.
- Audio-based learning sessions.
- Advanced semantic search across Sanskrit literature.
- Retrieval-Augmented Generation (RAG) for larger knowledge collections.
- Additional personas for Ayurveda, history, literature, and architecture.
- Admin dashboard for evidence management.
- User feedback and response ratings.
- Export conversations as PDF.
- Bookmarks and saved knowledge.
- Expert verification and evidence-contribution workflow.

16. Expected Outcome

Pragya AI aims to become a specialized AI knowledge platform that makes Indian Knowledge Systems more accessible, understandable, and relevant to modern users.

By combining specialized AI personas, multilingual interaction, image understanding, secure personal conversations, and a verified Sanskrit evidence database, the platform will demonstrate how traditional knowledge can be explored alongside modern technology without compromising factual accuracy.

Pragya AI is not intended to replace modern science with ancient knowledge. Its purpose is to create a meaningful bridge between the two through research, explanation, critical thinking, and responsible AI.